

Constructing and Solving Equations — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split. Where a question is worth more than one mark, award method marks even when the final answer is wrong — and take them away for a correct answer with no working only if the question said "show your working". The notes under each answer flag the specific mistake that question was built to catch.

Paper A — Medium

30 MARKS • NON-CALCULATOR

- 1** $x + 9 = 17 \rightarrow \mathbf{x = 8}$, check $8 + 9 = 17$ [6]
 $5x = 45 \rightarrow \mathbf{x = 9}$, check $5 \times 9 = 45$ [6]
 $3x + 2 = 20 \rightarrow \mathbf{x = 6}$, check $3(6) + 2 = 20$ [6]
One mark per box. Award the check box only if the substitution is actually shown — writing "correct" is not a check. This column exists to build the habit that makes the rest of the paper self-marking.
- 2** (a) $\mathbf{x = 17}$ [4]
 (b) $\mathbf{x = 8}$
 (c) $\mathbf{x = 24}$
 (d) $\mathbf{x = -8}$
(c) is the one students reverse — x was divided by 4, so multiply by 4. Answering 1.5 means they divided again. (d) has a negative solution, which is an ordinary answer and not a warning sign.
- 3** (a) $2x = 12$, $\mathbf{x = 6}$ [4]
 (b) $4x = 28$, $\mathbf{x = 7}$
 (c) $5x = 0$, $\mathbf{x = 0}$
 (d) $3x = 15$, $\mathbf{x = 5}$
(c) surprises people: $x = 0$ is a perfectly good solution, and $5(0) + 8 = 8$ checks out. Students who assume they have gone wrong and start again lose time they did not need to lose.
- 4** (a) Subtract $2x$: $3x = 18$ **[1]**, so $\mathbf{x = 6}$ **[1]** [4]
 (b) Subtract $3x$: $4x - 4 = 12$, then add 4: $4x = 16$ **[1]**, so $\mathbf{x = 4}$ **[1]**
Move the smaller x term, so that what is left stays positive. Both are checkable in seconds: $5(6) = 30 = 2(6) + 18$ $\square$ and $7(4) - 4 = 24 = 3(4) + 12$ $\square$
- 5** (a) Divide by 2: $x + 5 = 8$ **[1]**, so $\mathbf{x = 3}$ **[1]** [4]
 (b) Divide by 3: $x - 4 = 7$ **[1]**, so $\mathbf{x = 11}$ **[1]**
Accept expanding instead: $2x + 10 = 16$ and $3x - 12 = 21$ reach the same answers. What earns nothing is half-expanding to $2x + 5 = 16$, which is the error to watch for.
- 6** (a) $\mathbf{5n - 3 = 32}$ **[1]** [3]
 (b) $5n = 35$, so $\mathbf{n = 7}$ **[1]**
 (c) $7 \times 5 = 35$, and $35 - 3 = 32$ $\square$ **[1]**
The check in (c) must go back to the words, not to the student's own equation — otherwise a wrongly formed equation checks out perfectly and the error survives.
- 7** (a) $2(x + x + 3) = 26$, that is $\mathbf{4x + 6 = 26}$ **[1]** [3]
 (b) $4x = 20$, so $\mathbf{x = 5}$ **[1]**
 (c) Width **5 cm**, length **8 cm**; check $2(5 + 8) = 26$ $\square$ **[1]**
Stopping at $x = 5$ loses the third mark. The question asked about the rectangle; x was only the route there.

- 8** (a) $n + (n + 7) = 41$, that is $2n + 7 = 41$ [1] [2]
 (b) $2n = 34$, $n = 17$, so Ali has **17** and Ben has **24** [1]
Check both facts, not one: 24 is 7 more than 17 and $17 + 24 = 41$. A student who answers "17" alone has not answered the question asked.

Paper B — Medium

30 MARKS • NON-CALCULATOR

- 1** $x - 12 = 5 \rightarrow x = 17$, check $17 - 12 = 5$ [6]
 $8x = 72 \rightarrow x = 9$, check $8 \times 9 = 72$ [6]
 $2x + 7 = 19 \rightarrow x = 6$, check $2(6) + 7 = 19$ [6]
One mark per box, and the check must be shown. The first row is the one where students subtract when they should add — the 12 was taken away from x , so it goes back on.
- 2** (a) $x = -5$ [4]
 (b) $x = 9$
 (c) $x = 36$
 (d) $x = 5$
(d) has the x being subtracted. Add x to both sides to get $9 = 4 + x$. Answering -5 means the student subtracted 9 from 4 without dealing with the sign of the x first.
- 3** (a) $3x = 15$, $x = 5$ [4]
 (b) $5x = 40$, $x = 8$
 (c) $2x = -8$, $x = -4$
 (d) $4x = 28$, $x = 7$
(c) goes negative in the middle and stays there. Check it: $2(-4) + 15 = -8 + 15 = 7$. Students who reject a negative answer on sight will lose this mark every time it appears.
- 4** (a) Subtract $2x$: $4x - 5 = 15$, then add 5: $4x = 20$ [1], so $x = 5$ [1] [4]
 (b) Subtract $5x$: $3x + 3 = 21$, then subtract 3: $3x = 18$ [1], so $x = 6$ [1]
Checks: $6(5) - 5 = 25 = 2(5) + 15$ and $8(6) + 3 = 51 = 5(6) + 21$. Both sides coming out to the same number is the only confirmation anybody needs.
- 5** (a) Divide by 4: $x + 2 = 7$ [1], so $x = 5$ [1] [4]
 (b) Expand: $6x - 2 = 22$, so $6x = 24$ [1], and $x = 4$ [1]
In (b) dividing by 2 first also works and gives $3x - 1 = 11$ — accept either route. What does not work is $6x - 1 = 22$, where only the first term was multiplied.
- 6** (a) $6n + 5 = 41$ [1] [3]
 (b) $6n = 36$, so $n = 6$ [1]
 (c) $6 \times 6 = 36$, and $36 + 5 = 41$ [1]
Follow the sentence in order. A student who writes $6(n + 5) = 41$ has added before multiplying and will get a fraction — which is itself a signal that the equation was formed wrongly.
- 7** (a) $x + (x + 2) + (x + 4) = 30$, that is $3x + 6 = 30$ [1] [3]
 (b) $3x = 24$, so $x = 8$ [1]
 (c) Sides **8 cm, 10 cm and 12 cm**; check $8 + 10 + 12 = 30$ [1]
The perimeter of a triangle is just the three sides added — there is no doubling here, unlike the rectangle questions. Students who write $2(3x + 6) = 30$ have imported the wrong formula.
- 8** (a) $s + 4s = 45$, that is $5s = 45$ [1] [2]
 (b) $s = 9$, so the son is **9** and the father is **36** [1]
Letting s be the younger age keeps the algebra whole-numbered. Starting from the father gives $f + f/4 = 45$, which is correct but drags a fraction through the question for no reason.

Paper C — Hard

30 MARKS · NON-CALCULATOR

- 1** (a) $6x + 3 = 4x + 15$ **[1]**, so $2x = 12$ and $x = 6$ **[1]** [4]
 (b) $5x - 10 = 3x + 6$ **[1]**, so $2x = 16$ and $x = 8$ **[1]**
The first mark is for expanding correctly; the second for finishing. Checks: $3(13) = 39 = 4(6) + 15$ □ and $5(6) = 30 = 3(10)$ □
- 2** (a) $8x - 12 - 2x + 10 = 22$, so $6x - 2 = 22$ **[1]**, giving $x = 4$ **[1]** [4]
 (b) $3x + 12 - 2x + 2 = 20$, so $x + 14 = 20$ **[1]**, giving $x = 6$ **[1]**
Both hinge on the subtracted bracket: $-2 \times -5 = +10$ in (a) and $-2 \times -1 = +2$ in (b). A student who writes -10 and -2 gets 5 and 5, which look plausible and are both wrong.
- 3** (a) $x \div 4 = 5$, so $x = 20$ [4]
 (b) $x - 3 = 20$, so $x = 23$
 (c) $2x = 24$, so $x = 12$
 (d) $2x + 1 = 15$, so $x = 7$
(a) against (b) is the whole point: in (a) only the x is divided, so the $+3$ comes off first; in (b) the entire $(x - 3)$ is divided, so multiplying by 5 undoes it in one move. Reading (b) as $x \div 5 - 3$ gives 35 and is the standard error.
- 4** (a) $-2x = -6$ **[1]**, so $x = 3$ **[1]** [4]
 (b) Add $3x$ to both sides: $10 = 7x - 4$, so $14 = 7x$ **[1]**, giving $x = 2$ **[1]**
In (b), adding $3x$ rather than subtracting $4x$ is what keeps every coefficient positive. Accept the other route, but it produces $-7x = -14$ and one more chance to drop a sign.
- 5** (a) Gym A: **300 + 40m**. Gym B: **100 + 60m** **[1]** [4]
 (b) $300 + 40m = 100 + 60m$, so $200 = 20m$ **[1]**, giving $m = 10$ months **[1]**
 (c) At 15 months A costs 900 and B costs 1000, so **Gym A is cheaper by 100 rupees [1]**
The equation in (b) locates the crossover: below 10 months B's smaller joining fee wins, above 10 months A's cheaper rate wins. Students who answer (c) by guessing rather than substituting often pick the wrong side of the crossover.
- 6** (a) Angles on a straight line add to 180° , so $(2x + 10) + (3x - 5) + (x + 25) = 180$, that is **$6x + 30 = 180$ [1]** [5]
 (b) $6x = 150$ **[1]**, so $x = 25$ **[1]**
 (c) The angles are **60° , 70° and 50° [1]**; $60 + 70 + 50 = 180$ □ **[1]**
The angle fact is required in (a) and is a mark in itself. In (c) the final check is not optional — it is the only thing that distinguishes a correct answer from a plausible one, and it catches an arithmetic slip in the substitution.
- 7** (a) $n + (n + 1) + (n + 2) = 72$, that is **$3n + 3 = 72$ [1]** [5]
 (b) $3n = 69$, $n = 23$ **[1]**, so the numbers are **23, 24 and 25 [1]**
 (c) The sum is always $3n + 3 = 3(n + 1)$ **[1]**, which is 3 times a whole number and therefore always a multiple of 3. 70 is not a multiple of 3, so it is impossible **[1]**
In (c), "I tried some numbers and none worked" earns nothing — the argument must cover every case. The factorised form $3(n + 1)$ is what makes the claim provable rather than merely likely.

Paper D — Hard

30 MARKS · NON-CALCULATOR

- 1** (a) $6x - 12 = 4x + 4$ **[1]**, so $2x = 16$ and $x = 8$ **[1]** [4]
 (b) $6x + 15 - 4x + 4 = 27$, so $2x + 19 = 27$ **[1]**, giving $x = 4$ **[1]**
(b) is the subtracted bracket again: $-4 \times -1 = +4$. Check: $3(13) - 4(3) = 39 - 12 = 27$ □

- 2** (a) Multiply both sides by 4: $3x - 2 = 16$ [1], so $3x = 18$ and $x = 6$ [1] [4]
 (b) Multiply every term by 6: $3x + 2x = 60$ [1], so $5x = 60$ and $x = 12$ [1]
(b) is the one worth teaching from. Multiplying by 6 — the lowest common denominator of 2 and 3 — clears both fractions at once, and it must hit the 10 as well. Check: $6 + 4 = 10$ □
- 3** (a) $2((3x - 1) + (x + 2)) = 42$, so $2(4x + 1) = 42$, that is $8x + 2 = 42$ [2] [4]
 (b) $8x = 40$, so $x = 5$. Length **14 cm**, width **7 cm** [1]; check $2(14 + 7) = 42$ □ [1]
Simplify inside the bracket before doubling — students who double each side separately and then add usually lose a term. The check in (b) is worth a mark and catches a wrong x immediately.
- 4** (a) Plumber A: **250 + 180h**. Plumber B: **230h** [1] [4]
 (b) $250 + 180h = 230h$, so $250 = 50h$ [1], giving **h = 5 hours** [1]
 (c) At 3 hours A charges 790 and B charges 690, so **Plumber B is cheaper by 100 rupees** [1]
Plumber B has no fixed term at all, which students often mishandle by inventing one. Below 5 hours B wins; above 5 hours A's lower hourly rate overtakes the call-out fee.
- 5** (a) $n + (n+1) + (n+2) + (n+3) = 90$, so $4n + 6 = 90$, giving $n = 21$ [1]; the numbers are **21, 22, 23 and 24** [1] [4]
 (b) The sum is always $4n + 6 = 2(2n + 3)$ [1], which is 2 times a whole number and therefore always even [1]
Note what (b) does not claim: the sum is even but not necessarily a multiple of 4, since $2n + 3$ is always odd. A student who claims divisibility by 4 has over-read their own factorisation.
- 6** Student T moved the $2x$ across **without changing its sign**, adding $2x$ instead of subtracting it [1]. Subtracting $2x$ from both sides gives $3x - 3 = 12$, not $7x = 15$ [1] [5]
 Student U moved the -3 across **without changing its sign**, computing $12 - 3 = 9$ instead of $12 + 3 = 15$ [1]. Adding 3 to both sides gives $3x = 15$ [1]
 Correct: $3x = 15$, so $x = 5$ [1]
Both students made the same error — moving a term across without flipping its sign — one on an x term and one on a constant. Naming it once covers both, and it is the error that “doing it to both sides” is designed to make impossible.
- 7** (i) The mistake: dividing by 3 must divide **every** term, so the 6 becomes 2, not 6 [1]. [5]
 Correctly: $x + 2 = 6$, so $x = 4$ (or subtract 6 first: $3x = 12$) [1]
 (ii) The mistake: the 4 multiplies **both** terms in the bracket, so the 3 becomes 12, not 3 [1]. Correctly: $4x + 12 = 20$, so $4x = 8$ and $x = 2$ [1]
 The check: substitute the answer back into the **original** equation. $3(0) + 6 = 6$, not 18; and $4(4.25 + 3) = 29$, not 20 [1]
The final mark is the transferable one, and it is worth labouring: both students would have caught themselves in ten seconds. Note also that both errors are the same shape — a number applied to one term instead of all of them.